Can Frontier LLMs One-Shot Translate Lakota? (Mostly No.)

Evaluating Frontier LLM Translation Capability for Lakota

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No frontier LLM reliably translates Lakota.

And the standard surface metric (chrF++) hides how far short it falls.

6% → 60%

semantic equivalence across the seven models tested

best English → Lakota: 42.6 chrF++ (inadequate for use) · two LLM judges from different families agree, $\kappa = 0.75$ (0.89 weighted)

Lakȟótiyapi

the language under test

Why test this?

Lakȟótiyapi (Lakota) is a Siouan language UNESCO classifies as **critically endangered** — fewer than 2,000 first-language speakers, most over 65.

Today's LLMs advertise translation across dozens of languages, but training-data coverage is undisclosed. Users reasonably assume a supported language works. For an endangered language with scarce digitized text, that assumption is **untestable without direct evaluation.**

Setup

- **7 models.** Proprietary: Gemini 3.1 Pro, Claude Opus 4.6, Claude Sonnet 4.6, GPT-5.2. Open-weight: DeepSeek-V4-Pro, Qwen3.6-Plus, GLM-5.1.
- **200 sentence pairs** from the New Lakota Dictionary, conversational register, **both directions.**
- **Two conditions:** baseline vs. extended reasoning, where the API permits.
- **Metrics:** chrF++ and BLEU; diacritic-normalized chrF++; an LLM *semantic* judge on the English side.

Why two judges?

chrF++ rewards surface overlap; it cannot distinguish fluent-but-wrong from correct. We judge *meaning* with an LLM, English-to-English (no Lakota knowledge required). To rule out family-level self-judging bias, every pair is judged by **two models from different families** — Gemini 3 Flash and the independent Llama-3.3-70B — which agree substantially: Cohen's $\kappa = 0.75$ (0.89 quadratic-weighted).

Standard Lakota Orthography

SLO marks phonemic distinctions with diacritics that models often *drop or rearrange*:

- *ȟ* (caron) vs. *h* — distinct consonants
- *ą į ų* (ogonek) — nasalized vowels
- *á* (acute) — stress
- *ŋ* — velar nasal
- *'* — glottal stop

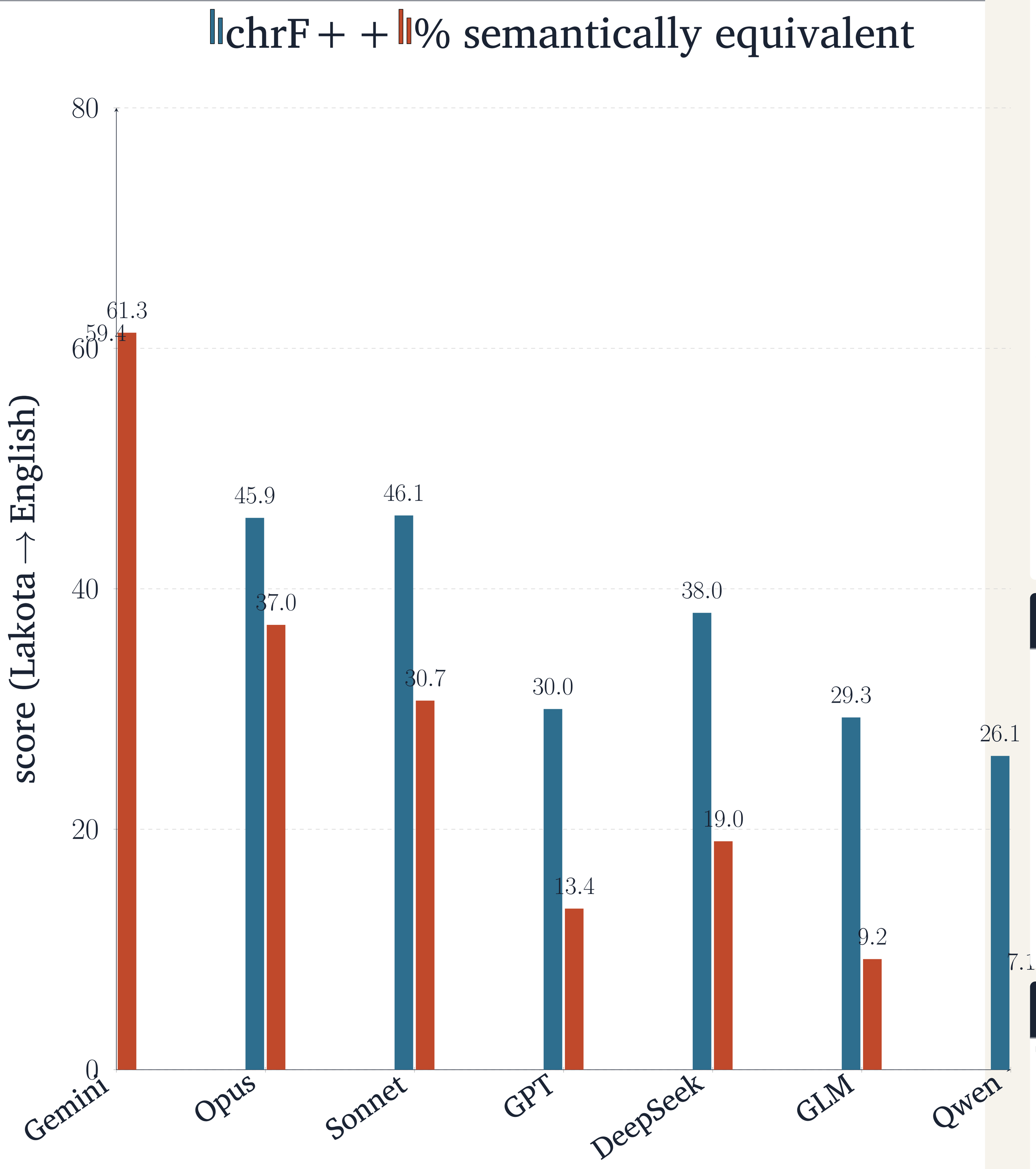
Stripping the marks adds **+3.3 to +6.5 chrF++** on English → Lakota — isolating orthographic from lexical error: models get the base characters roughly right and the marks inconsistent.

Translation quality (chrF++)

Model	L → E	E → L
Proprietary		
Gemini 3.1 Pro	**59.4**	**42.6**
Claude Opus 4.6	45.9	34.3
Claude Sonnet 4.6	46.1	27.0
GPT-5.2	30.0	23.6
Open-weight		
DeepSeek-V4-Pro	38.0	29.2
GLM-5.1	29.3	14.6
Qwen3.6-Plus	26.1	18.4

Best condition per model. Every model scores higher *out of* Lakota than *into* it; no open-weight model approaches Gemini.

Surface metrics overstate capability



chrF++ stays in the 30s–40s even where genuine semantic equivalence collapses below 15%. The two metrics agree for the strongest model and diverge sharply for the rest — so chrF++ alone is misleading for low-resource translation.

Semantic equivalence (L → E)

Model	Equiv.	Partial	Not
Gemini 3.1 Pro	**61.3**	23.1	15.6
Claude Opus 4.6	37.0	21.0	42.0
Claude Sonnet 4.6	30.7	23.8	45.5
DeepSeek-V4-Pro	19.0	14.0	67.0
GPT-5.2†	13.4	9.8	76.8
GLM-5.1	9.2	6.5	84.2
Qwen3.6-Plus	7.1	7.6	85.4

% of judged pairs, reasoning condition. † GPT-5.2 baseline = 6.0% / reasoning = 13.4%; the headline "6%" is the baseline floor.

How models fail

Reference	Model output	chrF++
We live in a small trailer house with no electricity.	"small *sacred* tipi"	18.6
Make coffee (*Wakȟálya yo*).	"Sanctify it!"	2.9
Did my daughter call?	"Did my younger brother come?"	27.6
Do you have a car?	"Do you have a lighter?"	65.9
I want to talk to you.	"I will be very happy."	6.5

Idioms and coined terms are translated *element by element*: *Wakȟálya yo* ("make coffee," literally "make it boil") becomes "sanctify it." The one-word near-miss (65.9) outscores output unrelated to the source — surface overlap and meaning preservation diverge.

Reasoning buys calibration, not capability

For the open-weight models, extended reasoning leaves translation quality and meaning *flat* but raises **refusals**:

- Qwen3.6-Plus, E → L: **0** → **36** refusals (of 200) when reasoning is enabled.
- GLM-5.1, L → E: **1** → **16**.

Reasoning surfaces the limitation rather than overcoming it. On a language this far outside the training distribution, the binding constraint is **knowledge, not inference** — reasoning's contribution is calibration, not capability.

Takeaways

- No frontier LLM reliably translates Lakota in either direction (as of 2026).
- **chrF++ overstates** low-resource capability — pair it with an LLM semantic judge.
- Open-weight models do not close the gap; reasoning changes *refusals*, not quality.

Origin

This evaluation began as a response to a provocation from Benjamin Bratton (UCSD): *"just vibe-code a Lakota translation app over the weekend."* We built the evaluation that would test whether such an app could exist.

Data & code

github.com/robotson/lakota-translation-benchmark

Evaluation corpus withheld (community data); the repo holds scripts, aggregate results, and the paper.